

LAB 3 — COMPONENT MODELLING & ARCHITECTURAL PATTERN SELECTION

Self-Service Coffee Kiosk | Layered Architecture

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Architecture Selection

We chose Layered Architecture for the Self-Service Coffee Kiosk system.

Architectural Choice & Reasons

- 1. Clear separation of concerns:** the User Interface handles touch-screen interaction, the Order Manager handles ordering logic, and the Database handles menu, pricing and order data. This makes the system easier to understand, maintain and test.
- 2. Suitable for a focused kiosk application:** the system has a small, well-defined feature set and does not require independently deployed services. A layered design keeps the architecture simple while supporting the required payment and receipt workflows.

Security Advantage

The layered structure prevents the touch-screen interface from directly accessing stored data. Payment requests are handled through the Payment Service, while database access is kept behind the Order Manager, reducing unnecessary exposure of sensitive operations.

Performance Benefit

For a single kiosk, the layered design avoids the additional network and operational overhead of a microservices-based solution. Menu and pricing access remains centralized, while order processing follows a short, predictable path through the application layers.

Components & Interfaces

The diagram contains five components: User Interface, Order Manager, Payment Service, Receipt Printer and Database. It shows four required service connections: Order API, Payment API, Data Access and Print API, using provided/required interface notation and labelled data flows.